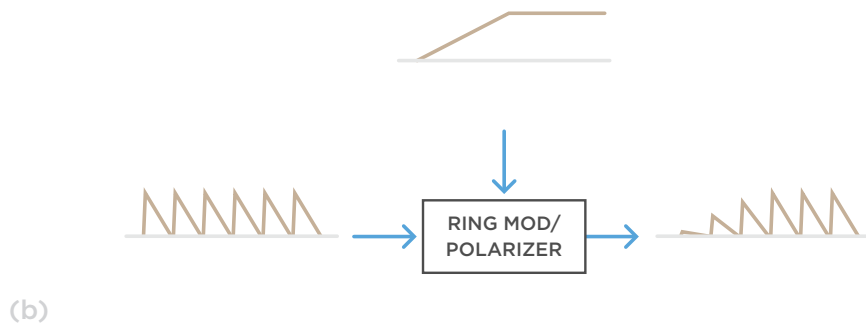
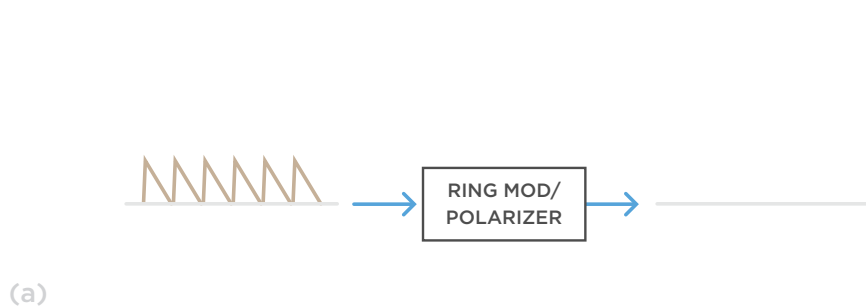


VCA Alternatives

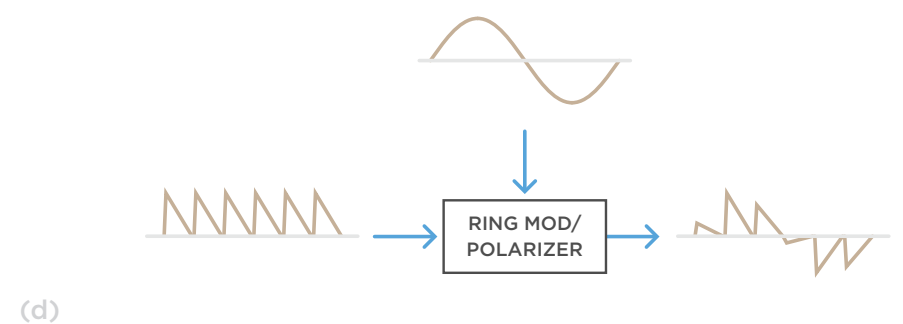
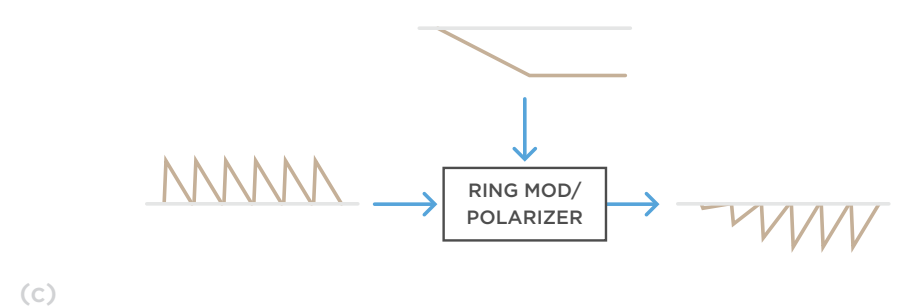
Ring modulator / Polarizer

- Audio
- Control Voltage
- Trigger / Gate



a - Polarizers and ring modulators work as bipolar VCAs. If you send them a signal without any modulation, for example a uni-polar looping envelope, the incoming signal is blocked.

b - If you modulate them with a positive signal, the incoming signal is amplified in a regular way.

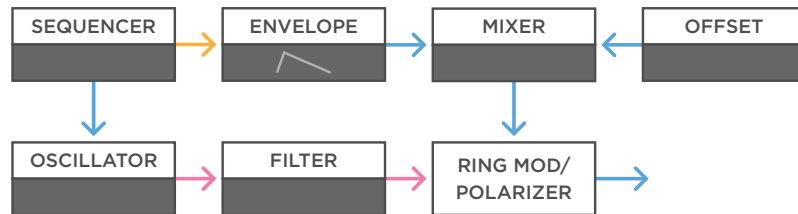
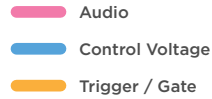


c - If modulate them with a negative signal, the incoming signal is amplified in an inverted way.

d - And if you modulate them with a bipolar signal, like a slow sine wave LFO, the input signal slowly sweeps between positive and negative.

VCA Alternatives

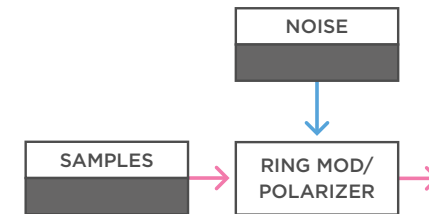
Ring modulator / Polarizer



(a)

a - So, if you make a standard synth voice with an oscillator and filter, you can use a polarizer or ring mod as a closed VCA. Then just send a gate from a keyboard or sequencer to a regular positive envelope and modulate the polarizer like you would a normal VCA. If you like, you can mix the signal of the envelope with offset, so you have manual control over the VCA, as well the amount of influence from the envelope.

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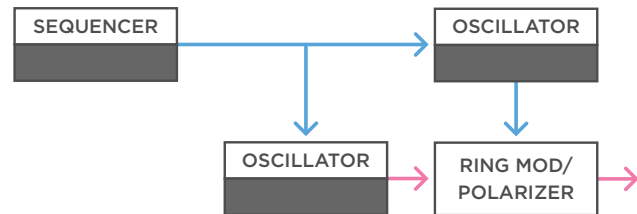
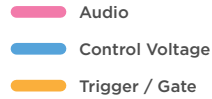


(b)

b - Used as a ring-modulator you can get interesting audible effects, by sending the module audio rate signals, and modulating that audio signal with a bipolar control voltage. Preferably also audio rate signals. For example you can feed it a drum loop, and add dirt and dust to it by modulating the ring-modulator with noise.

VCA Alternatives

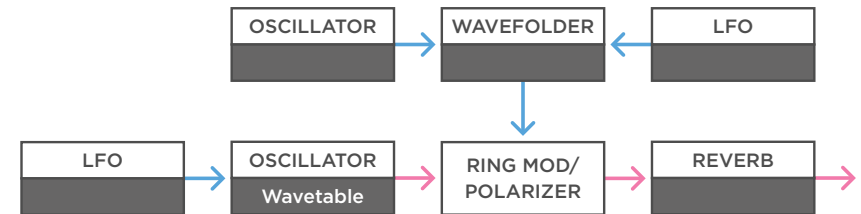
Ring modulator / Polarizer



(a)

a - You can create more classic ring-mod sounds, by using two harmonically tuned oscillators in a melodic patch.

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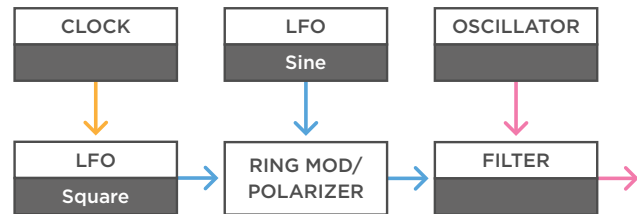
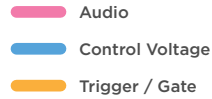


(b)

b - But using ring modulators in drone patches, using two untuned oscillators with more complex modulated wave shapes or folding possibilities, can also create lovely drone sounds and textures. Just add a lot of reverb!

VCA Alternatives

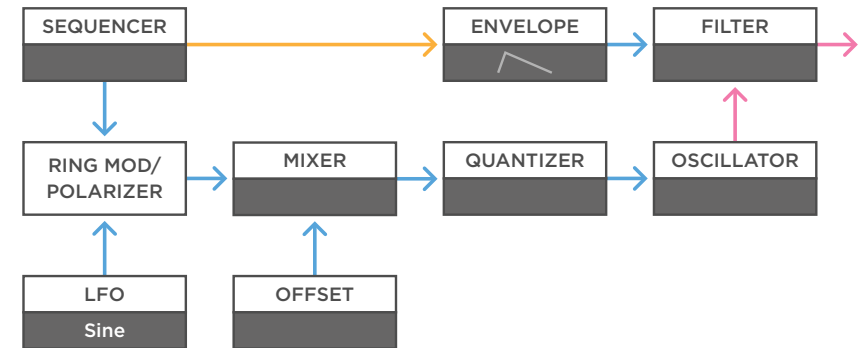
Ring modulator / Polarizer



(a)

a - Used as a Polarizer you can invert incoming bipolar modulation signals over time. For example, if feed the polarizer a tempo synced square wave LFO, and modulate it with a slow sine wave LFO, you can create interesting rhythmic variations.

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(b)

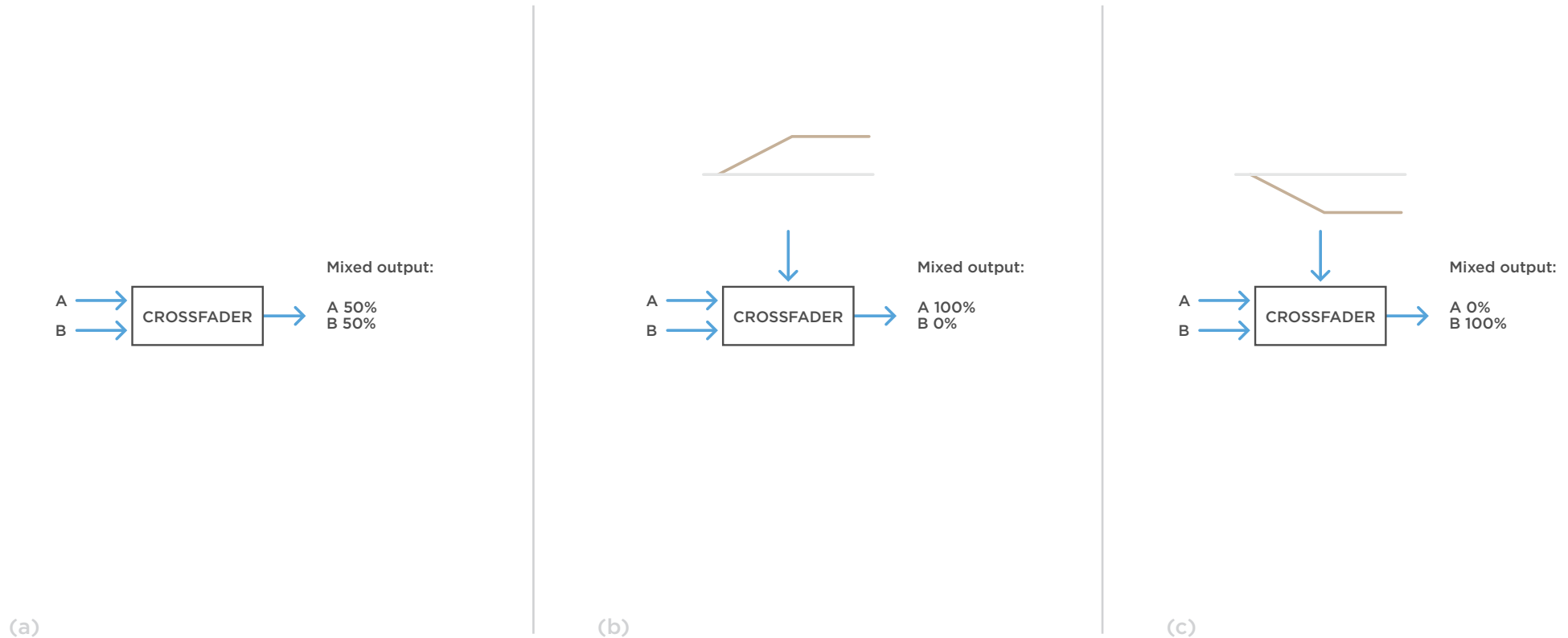
b - In this example a simple 3 step sequence is send through a polarizer, and then mixed with some offset and quantized again. The resulting 1v/oct signal is used to create a simple melodic voice. If you use something like a slow Sinewave LFO on the polarizer, you can sweep that sequence from a regular to an inverted sequence.

VCA Alternatives

Crossfader

- Audio
- Control Voltage
- Trigger / Gate

MONOTRAIL TECH TALK



a - Another creative tool that pretty easily doubles as a regular VCA, is a crossfader. The concept of a crossfader is simple. You can send it two different signals, let's call them A and B. The module outputs a mix of those signals. A common setup for crossfaders is to output a 50/50% mix of the incoming signals, when the module isn't modulated.

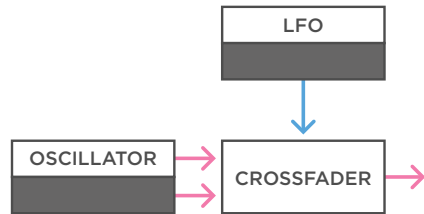
b - However, by sending the module a positive control voltage, you can shift the output balance to a 100% signal A and 0% signal B.

c - And if you modulate the module with a negative voltage to 0% signal A and 100% signal B.

VCA Alternatives

Crossfader

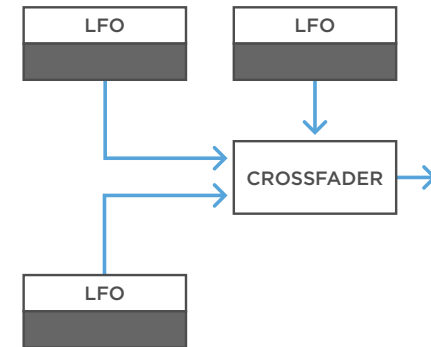
- Audio
- Control Voltage
- Trigger / Gate



(a)

a - You can use crossfaders for all sorts of patches. For example, feed two waveforms of the same oscillator into a crossfader, and use a slow LFO to create analog wave morphing.

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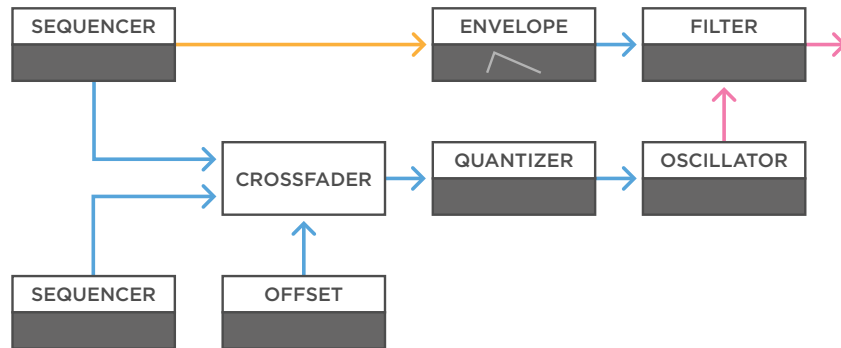
(b)

b - This same trick also works when you feed the crossfader different waveforms from a single, or multiple LFOs.

VCA Alternatives

Crossfader

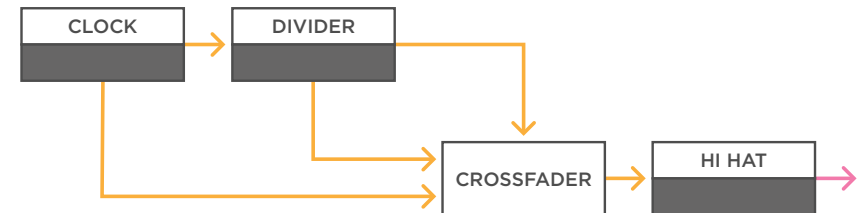
Audio
Control Voltage
Trigger / Gate



(a)

a - In this example, two 8 step sequences are sent to the inputs of the crossfader. The mix output is sent to a quantizer, and controlling the pitch of an oscillator in a simple voice. By sending the crossfader an offset voltage, you can manually crossfade between the two melodic patterns, creating new sequences in between.

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(b)

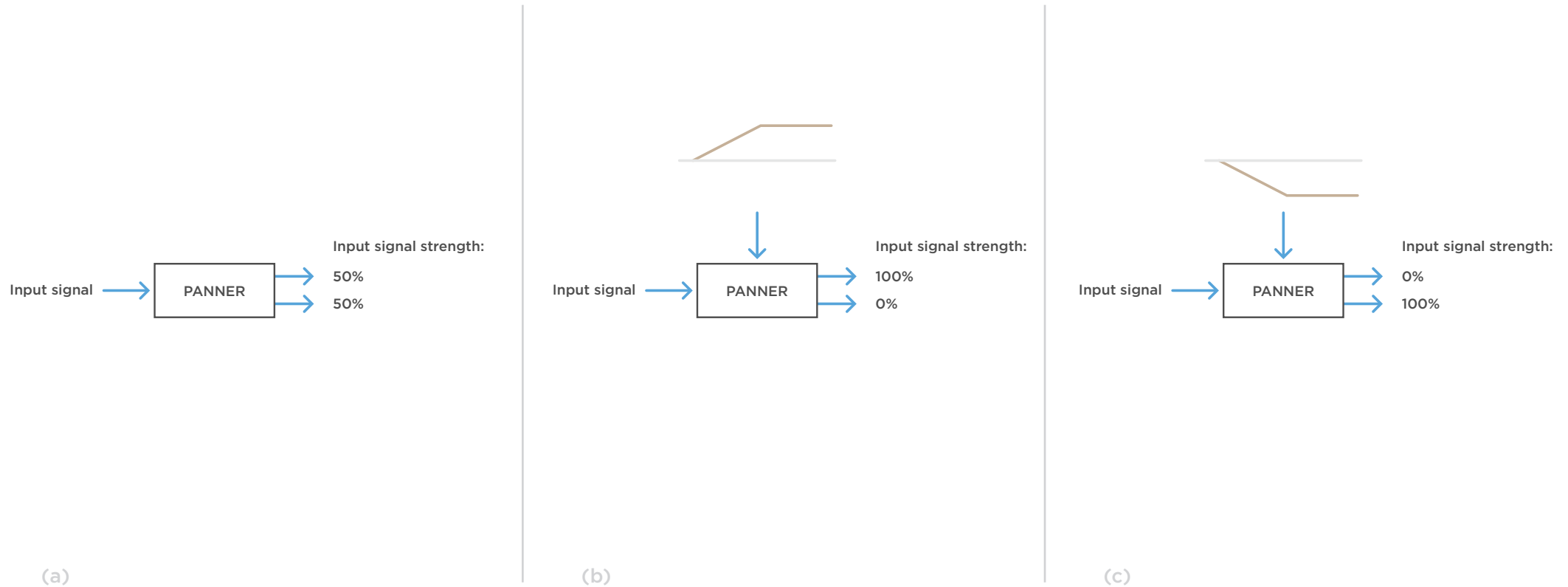
b - Crossfaders can also be used as voltage controlled two input switches. For example use a fast clock as input A, a slower division as input B, and an even slower division to change the balance.

VCA Alternatives

Panner

- Audio
- Control Voltage
- Trigger / Gate

MONOTRAIL TECH TALK



a - Then there's the voltage controlled panner. Another module that can be used as a VCA for audio and control voltages, as well as a lot of creative patches. The function is similar to the crossfader, but reversed. It only has one input, but two outputs.

a - The incoming signal is divided 50/50% when the module is not modulated.

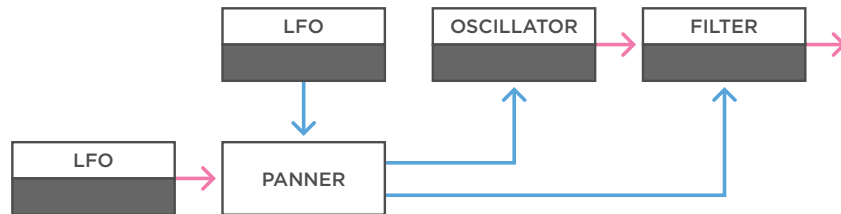
b - Positive CV signals can route the incoming signal entirely to output A.

c - And negative signals can route the incoming signal entirely to output B.

VCA Alternatives

Panner

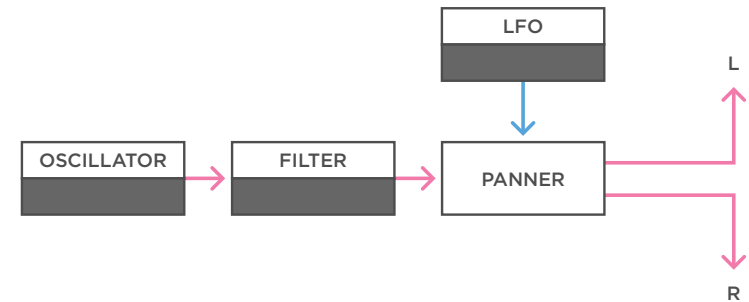
Audio
Control Voltage
Trigger / Gate



(a)

a - This allows for a lot of create fun. For example, use it to divide an audio rate LFO, between modulating the frequency of the filter, and an oscillator in a single voice. When you use a slow sinewave LFO to sweep the signal between the two destinations, you create interesting textures.

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(b)

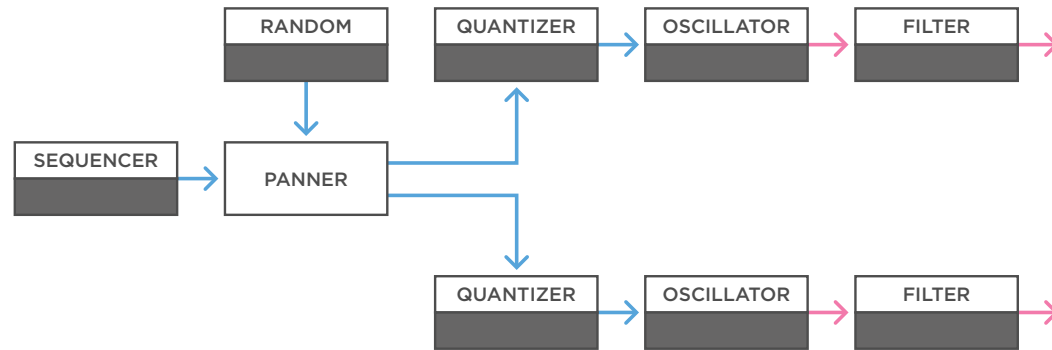
b - Of course, you can also use panners to turn a monophonic sound, like a drum loop or synth voice into a stereo sound. You can either send the panner slow changing signals like an LFO to sweep the audio left and right, or audio rate signals, to create fat sounds with a wide stereo field.

VCA Alternatives

Panner

- Audio
- Control Voltage
- Trigger / Gate

MONOTRAIL TECH TALK



(a)

a - Or, to give another melodic example, use a single 3 step sequence, and use something like an LFO or smooth random voltage to divide that sequence over a dual quantizer, controlling the pitch of two different voices.